Homework 4

Problem 3:

a) Code in folder.

b) ModelSim simulation:



c) The time it takes to complete one operation would be 64 gate-delay units. Each CAS module contains 5 full adders connected in a ripple carry configuration, this means that the carry propagation through the CAS determines the critical path. Each full adder requires 2 units for the sum output and 3 units for the carry out. The first CAS module contributes 15 units, while each of the remaining three CAS modules contributes 16 units due to the additional control logic. Adding the delays gives a grand total of: 15+16+16+16 = 63 units. The final quotient correction adds 1 additional unit, resulting in a total delay of 64 units.